HIMAKESH KRISHNA

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PROFESSIONAL SUMMARY

Data Science practitioner (IIT Guwahati PG Certification, 2025–2026) with hands-on experience building end-to-end ML pipelines across NLP, computer vision, and unsupervised learning — including a 95% accuracy BBC News classifier, a 95.52% CIFAR-10 model with Grad-CAM explainability, and a customer segmentation dashboard deployed on Streamlit. Six years of professional motion graphics and graphic design experience informs a rare strength in data visualization and communicating complex insights to non-technical stakeholders.

TECHNICAL SKILLS

Languages: Python, SQL, HTML/CSS, JSON,

ML / DL Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch. XGBoost, Random Forest, Gradient Boosting, Stacking Ensembles

NLP: NLTK, Word2Vec, HuggingFace (T5, BERT), BiLSTM, Seq2Seq, Bahdanau Attention

Computer Vision: CNN, EfficientNetV2M, DenseNet121, Transfer Learning, Grad-CAM

Unsupervised ML: K-Means, Gaussian Mixture Models, Agglomerative Clustering, DBSCAN, PCA

Data Tools: Pandas, NumPy, Matplotlib, Seaborn, SHAP, SMOTE, Jupyter Notebook

Deployment & Version Control: Streamlit, Git / GitHub

Other: Feature Engineering, EDA, Statistical Analysis, Class Imbalance Handling, Predictive Modeling, A/B Testing, Hypothesis Testing

PROJECTS

BBC News Multi-Class NLP Classification & Summarization System *2025*

Stack: Python · Logistic Regression · BiLSTM · Word2Vec · T5-small · HuggingFace · Keras · Streamlit |

- Built an end-to-end NLP pipeline on **2,225 BBC News articles**; Logistic Regression achieved **95% accuracy** and Macro F1 of **0.87** across 5 categories (Technology, Politics, Business, Sports, Entertainment).
- Resolved severe class imbalance via class-weighting, improving minority Technology-class Recall from **12% to 64%** (433% improvement) in the BiLSTM model.
- Fine-tuned **T5-small** for abstractive headline generation and implemented **Seq2Seq** model — spanning classical to transformer-era NLP in a single pipeline.

CIFAR-10 Computer Vision Pipeline with Grad-CAM Explainability *2025*

Stack: TensorFlow · Keras · EfficientNetV2M · DenseNet121 · Custom CNN · Grad-CAM · Streamlit |

- Benchmarked 3 CNN architectures (Custom CNN, DenseNet121, EfficientNetV2M) on **CIFAR-10 (60K images, 10 classes)**; identified and fixed a double-preprocessing normalization bug suppressing accuracy by ~10%.
- Trained EfficientNetV2M with **two-phase fine-tuning** (feature extraction + gradual unfreeze), achieving **95.52% test accuracy** — 6% above custom CNN baseline.
- Implemented **Grad-CAM explainability** for nested Functional API model enabling visual inspection of model attention across all 10 classes.

Credit Card Customer Behavioral Segmentation *2025*

Stack: Python · K-Means · GMM · Agglomerative · DBSCAN · PCA · Scikit-learn · Pandas · Streamlit |

- Engineered a customer segmentation pipeline on **8,950 credit card customers**; applied winsorization, log-transforms, and StandardScaler normalization across 17 raw features.
- Applied **PCA retaining ~96% variance** to reduce dimensionality; benchmarked K-Means, GMM, Agglomerative, and DBSCAN via silhouette score and information criteria.

- Identified 4 actionable customer personas; deployed as **3-page Streamlit dashboard** for stakeholder-ready business insights.

EDUCATION

Indian Institute of Technology (IIT), Guwahati *Aug 2025 – Jun 2026*

Post-Graduate Certification in Data Science | Guwahati, Assam

- Relevant coursework: Machine Learning, Deep Learning, NLP, Statistics & EDA, Predictive Modeling, Data Visualization
- Certifications: IITG Statistics & EDA | IITG Introduction & Programming in Data Science

WORK EXPERIENCE

Nerul Marketing *Dec 2024 – Dec 2025*

Graphic Designer & Motion Graphics Specialist | Remote

- Drove **400% revenue growth** through data-informed creative strategy and systematic client communication across high-value enterprise accounts.
- Maintained **~90% first-pass client approval rate** by developing a client-perspective translation framework — directly applicable to communicating ML model insights to non-technical stakeholders.
- Grew from junior designer to **lead advisor within 12 months**; founder delegated strategic creative direction to me and regularly sought input on project approach — led the designer team on client deliverables end-to-end.

CERTIFICATIONS

- **IITG Advanced AI: Transformers, LLMs & Generative Intelligence** — Indian Institute of Technology, Guwahati
- **IITG Deep Learning & Neural Architectures** — Indian Institute of Technology, Guwahati
- **IITG Applied ML & Pattern Discovery** — Indian Institute of Technology, Guwahati
- **IITG Introduction & Programming in Data Science** — Indian Institute of Technology, Guwahati